wrf2ommuu

May 15, 2025

[1]:
```
#1.        Import libraries such as Pandas, matplotlib, NumPy, and seaborn and␣
↪load the insurance dataset

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('insurance.csv')
print(df)
```

```
      age     sex     bmi  children smoker     region      charges
0      19  female  27.900         0    yes  southwest  16884.92400
1      18    male  33.770         1     no  southeast   1725.55230
2      28    male  33.000         3     no  southeast   4449.46200
3      33    male  22.705         0     no  northwest  21984.47061
4      32    male  28.880         0     no  northwest   3866.85520
…   …      …      …       …    …         …          …
1333   50    male  30.970         3     no  northwest  10600.54830
1334   18  female  31.920         0     no  northeast   2205.98080
1335   18  female  36.850         0     no  southeast   1629.83350
1336   21  female  25.800         0     no  southwest   2007.94500
1337   61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

- 2.Check the shape of the data along with the data types of the column

[2]:
```
print(df.shape)
```

```
(1338, 7)
```

[3]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
```

```
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
None
```

[4]:
```
df.dtypes
```

[4]:
```
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
charges     float64
dtype: object
```

- 3.Check missing values in the dataset and find the appropriate measures to fill in the missing values

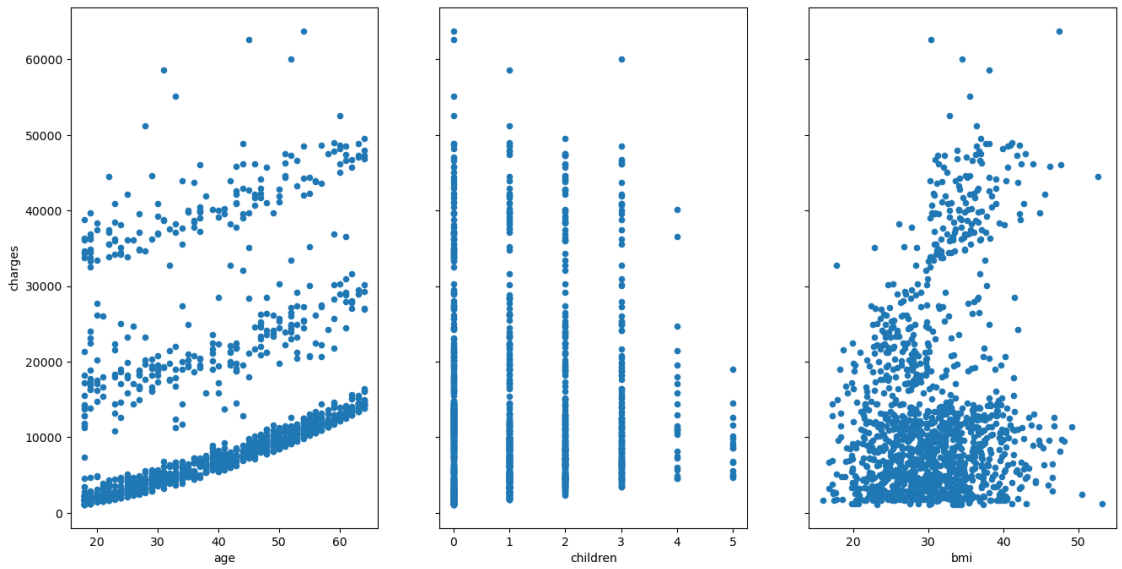
[5]:
```
df.isna().sum()
```

[5]:
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

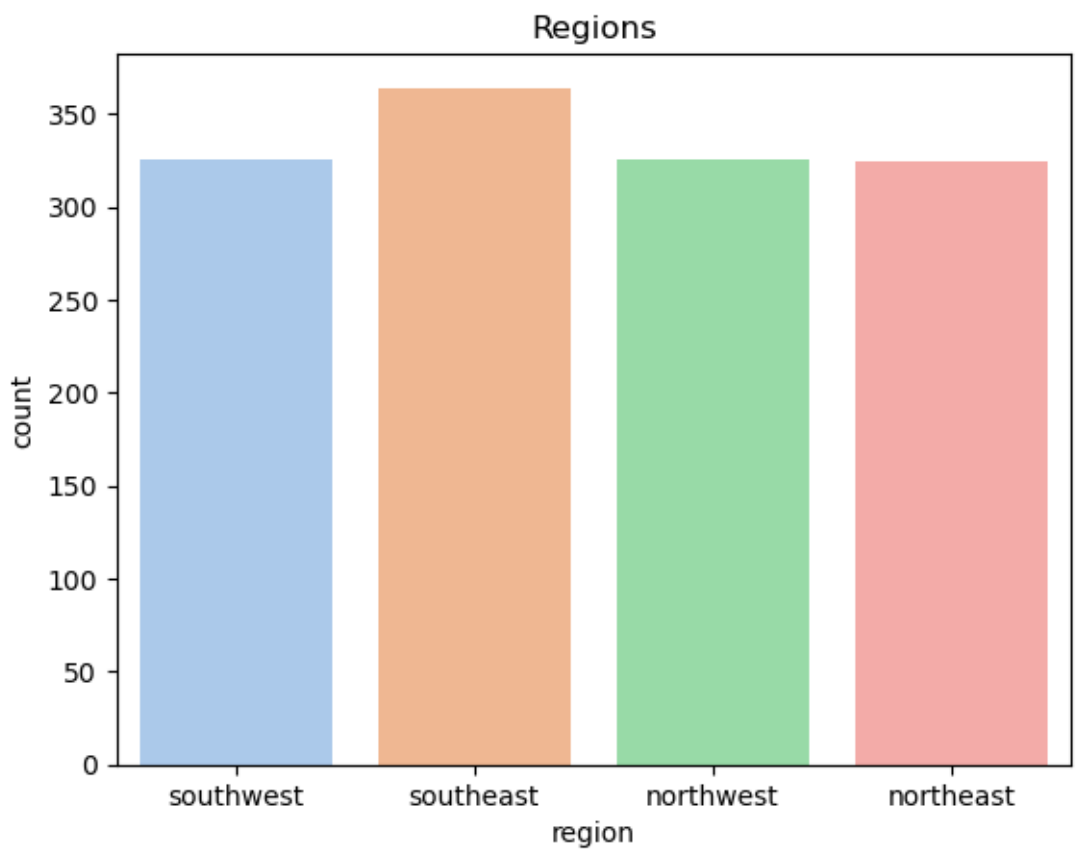
- 4.Explore the relationship between the feature and target column using a count plot of categorical columns and a scatter plot of numerical columns

[6]:
```
fig,axs =plt.subplots(1,3,sharey=True)
df.plot(kind='scatter',x='age',y='charges',ax=axs[0],figsize=(16,8))
df.plot(kind='scatter',x='children',y='charges',ax=axs[1])
df.plot(kind='scatter',x='bmi',y='charges',ax=axs[2])
```

[6]:
```
<Axes: xlabel='bmi', ylabel='charges'>
```

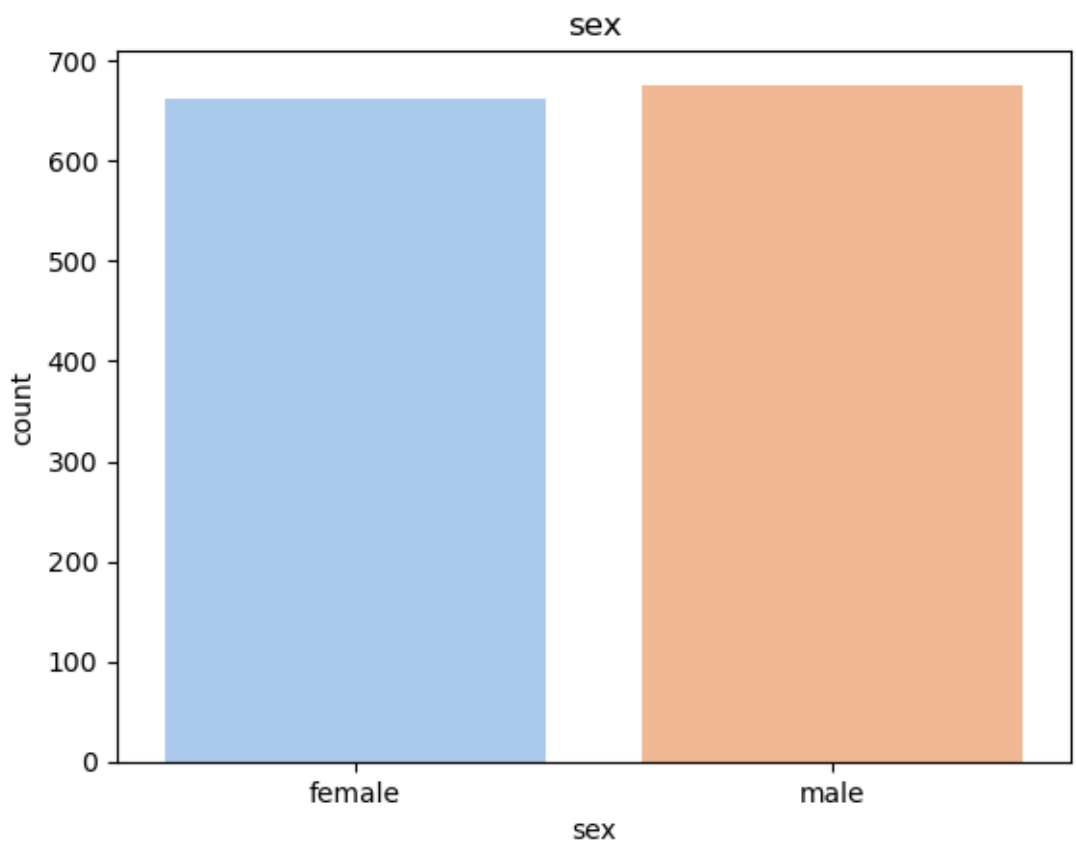


[7]:
```
sns.countplot(data=df,x='region',palette='pastel')
plt.title('Regions')
plt.show()
```

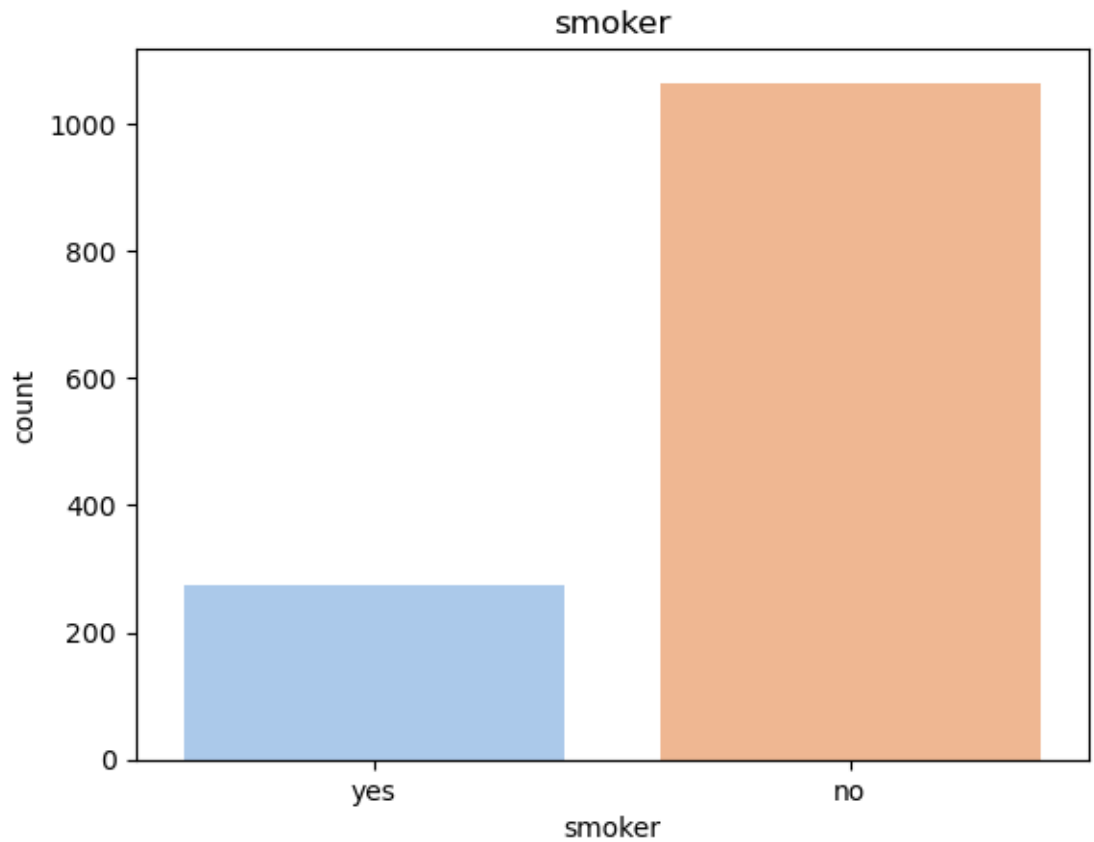


[8]:
```
sns.countplot(data=df,x='sex',palette='pastel')
plt.title('sex')
```

[8]: Text(0.5, 1.0, 'sex')

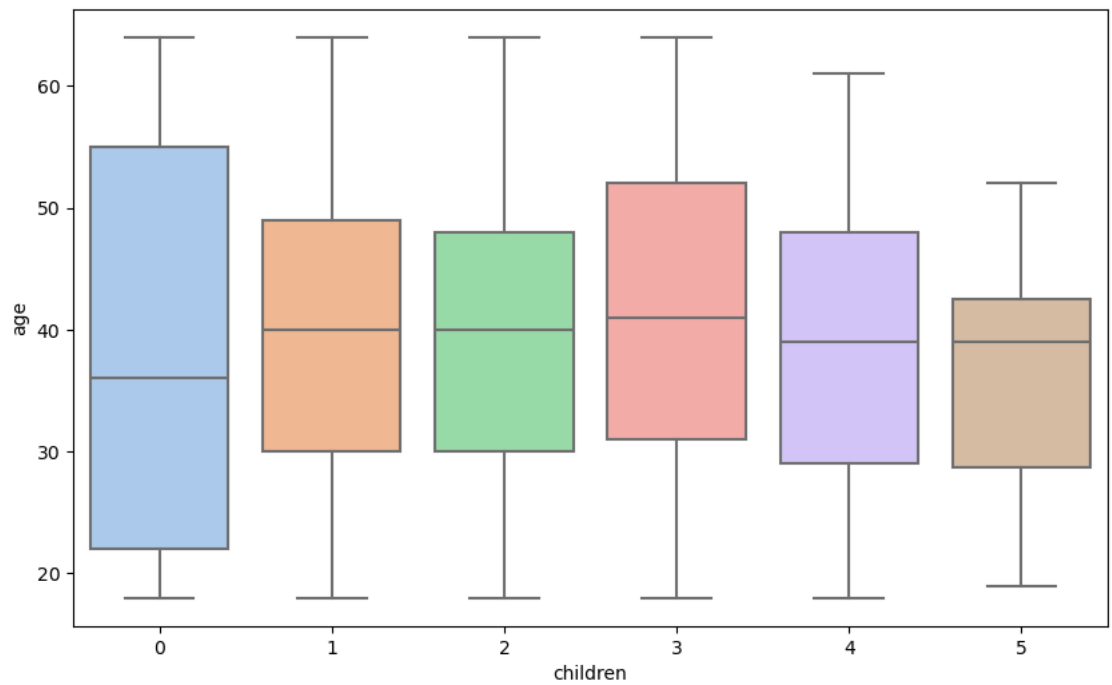


[9]:
```
sns.countplot(data=df,x='smoker',palette='pastel')
plt.title('smoker')
plt.show()
```

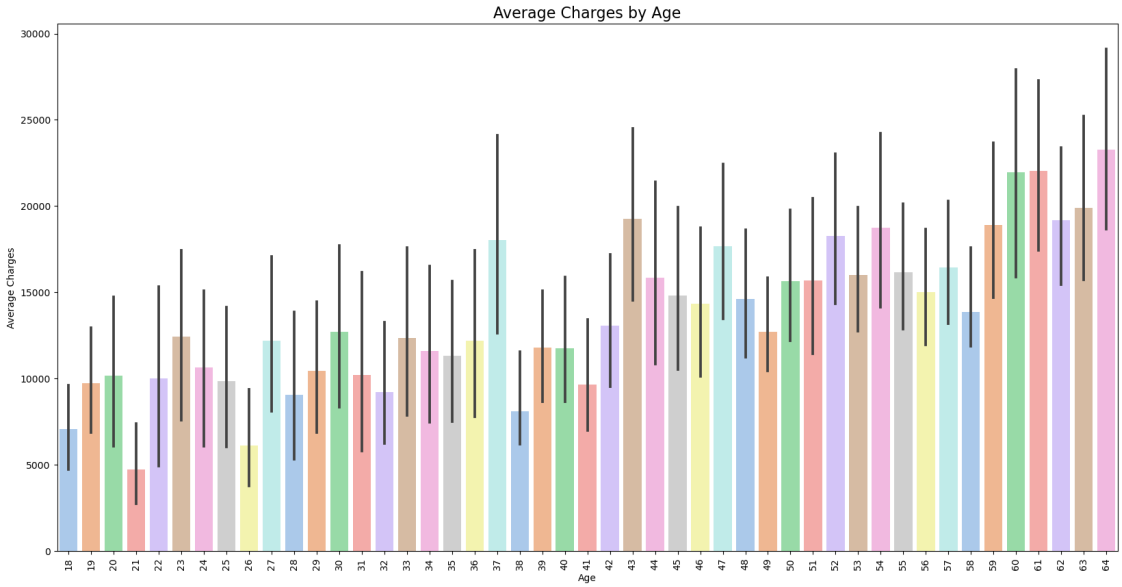


- 5.Perform data visualization using plots of feature vs feature

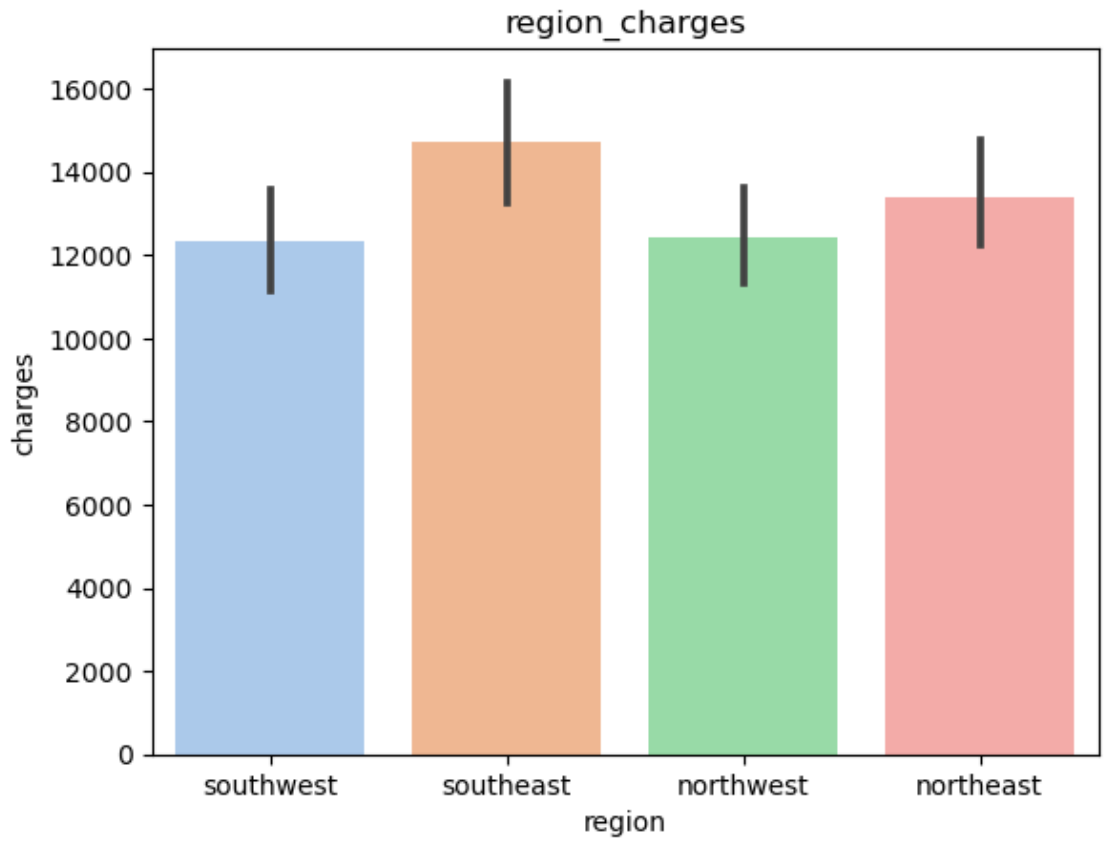
[10]:
```
plt.figure(figsize=(10,6))
sns.boxplot(x='children',y='age',data=df,palette='pastel')
plt.show()
```



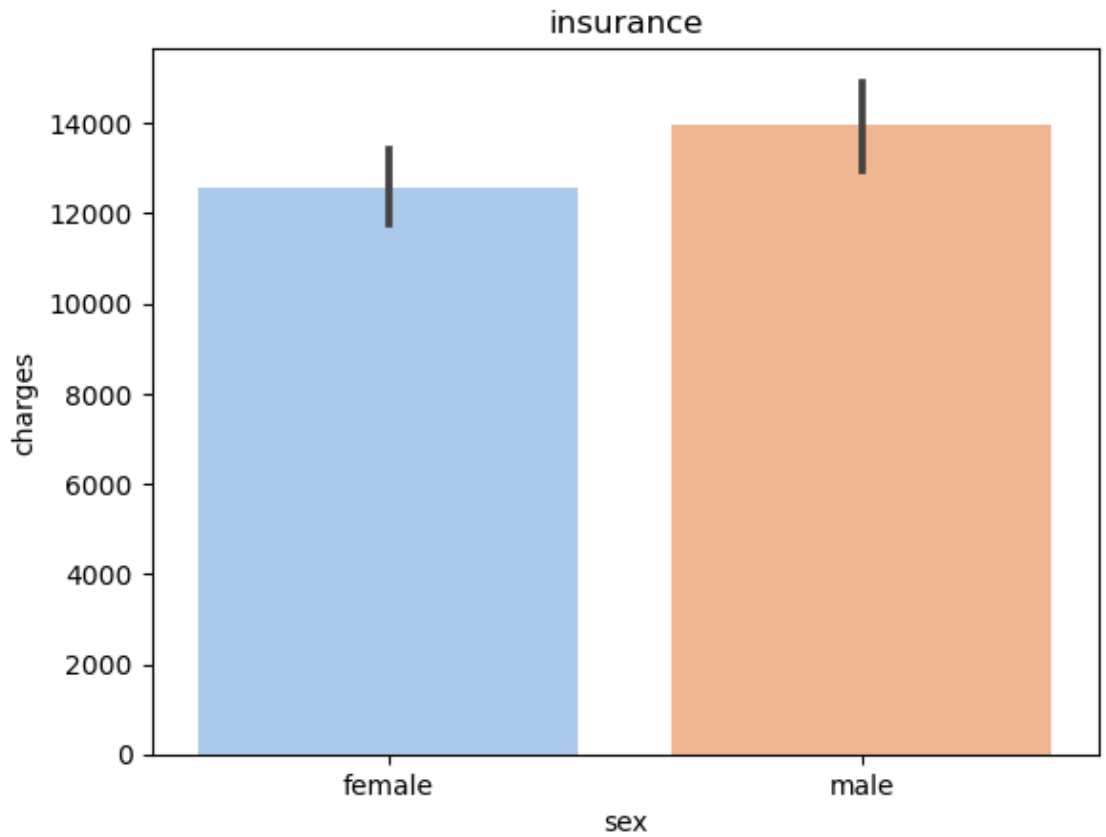
[18]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(20, 10))
sns.barplot(x='age', y='charges', data=df, palette='pastel')
plt.title('Average Charges by Age', fontsize=16)
plt.xlabel('Age')
plt.ylabel('Average Charges')
plt.xticks(rotation=90)
plt.show()
```



[35]:
```
#Draw a bar plot for region vs charge
import matplotlib.pyplot as plt
import seaborn as sns

sns.barplot(x='region',y='charges',data=df,palette='pastel')
plt.xlabel('region')
plt.ylabel('charges')
plt.title('region_charges')
plt.show()
```

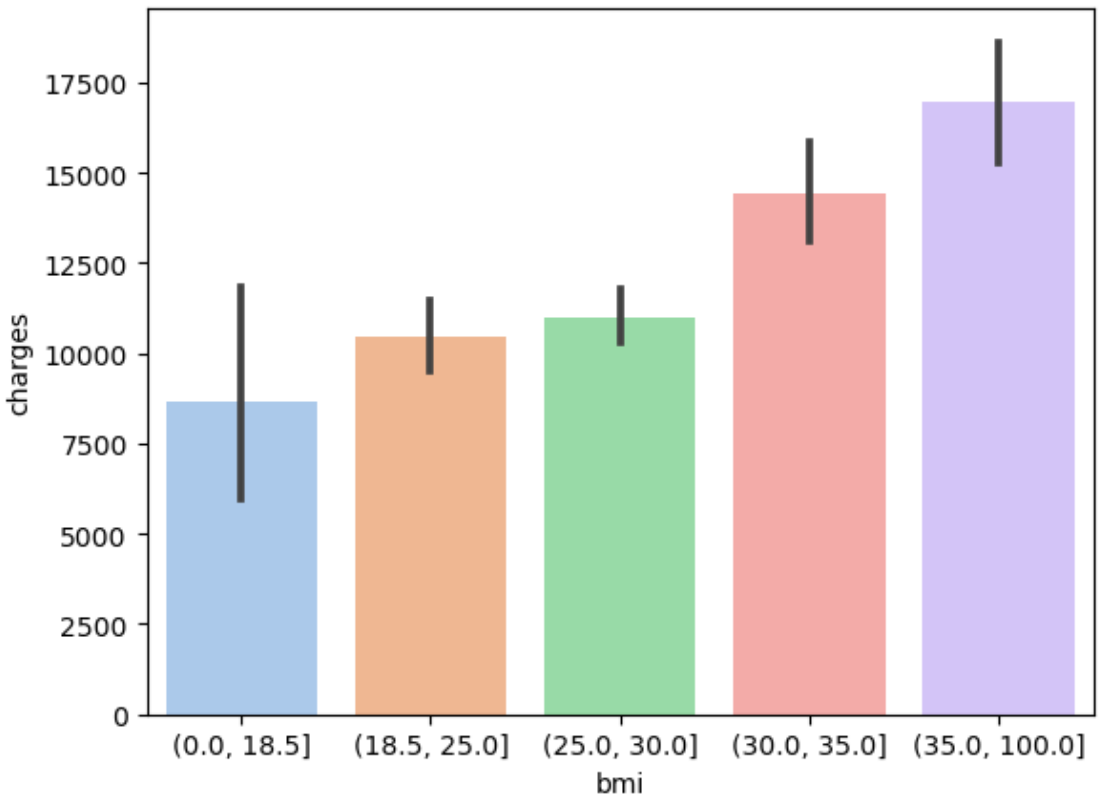


[34]:
```
#Draw a bar plot for sex vs charges
import matplotlib.pyplot as plt
import seaborn as sns

sns.barplot(x='sex',y='charges',data=df,palette='pastel')
plt.xlabel('sex')
plt.ylabel('charges')
plt.title('insurance')
plt.show()
```

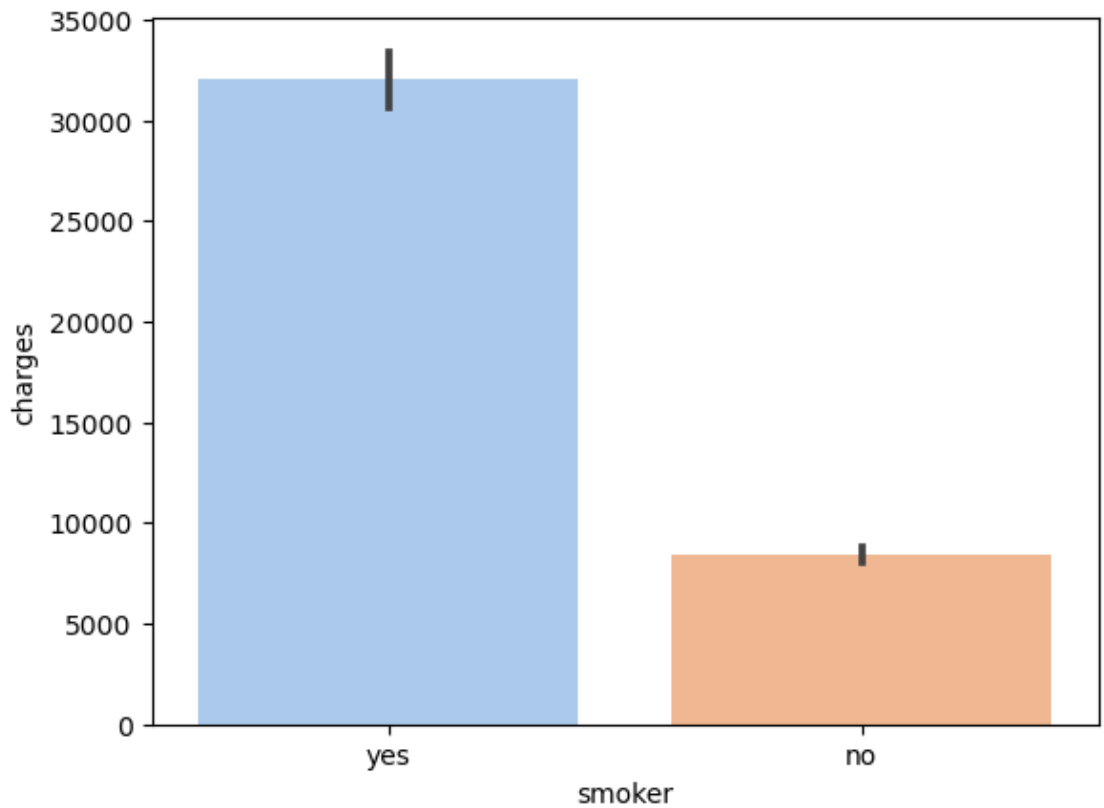


[43]:
```
# Draw a barplot for bmi vs charge
import matplotlib.pyplot as py
import seaborn as sns

bins=[0,18.5,25,30,35,100]
#labels=['Underweight', 'Normal', 'Overweight', 'Obese I', 'Obese II+']
df['bmi_group']=pd.cut(df['bmi'],bins=bins)

sns.barplot(x='bmi_group',y='charges',data=df,palette='pastel')
plt.xlabel('bmi')
plt.ylabel('charges')
plt.show()
```

```
C:\Users\swath\anaconda3\Lib\site-packages\seaborn\categorical.py:641:
FutureWarning: The default of observed=False is deprecated and will be changed
to True in a future version of pandas. Pass observed=False to retain current
behavior or observed=True to adopt the future default and silence this warning.
  grouped_vals = vals.groupby(grouper)
```

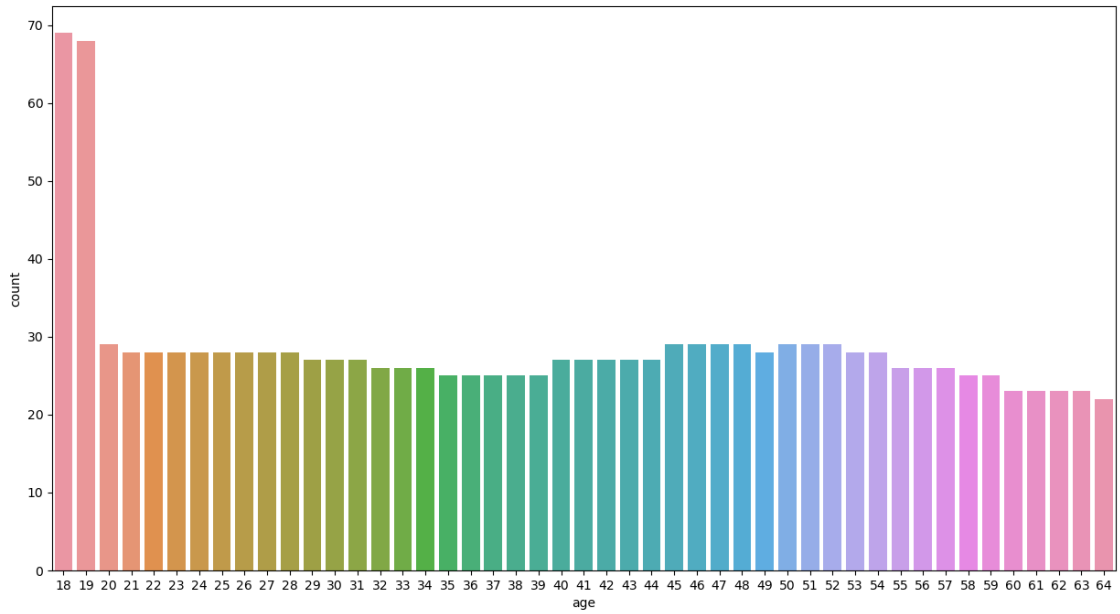


[45]:
```
# Draw a barplot for smoker vs charge
import matplotlib.pyplot as plt
import seaborn as sns

sns.barplot(x='smoker',y='charges',data=df,palette='pastel')
plt.xlabel('smoker')
plt.ylabel('charges')
plt.show()
```

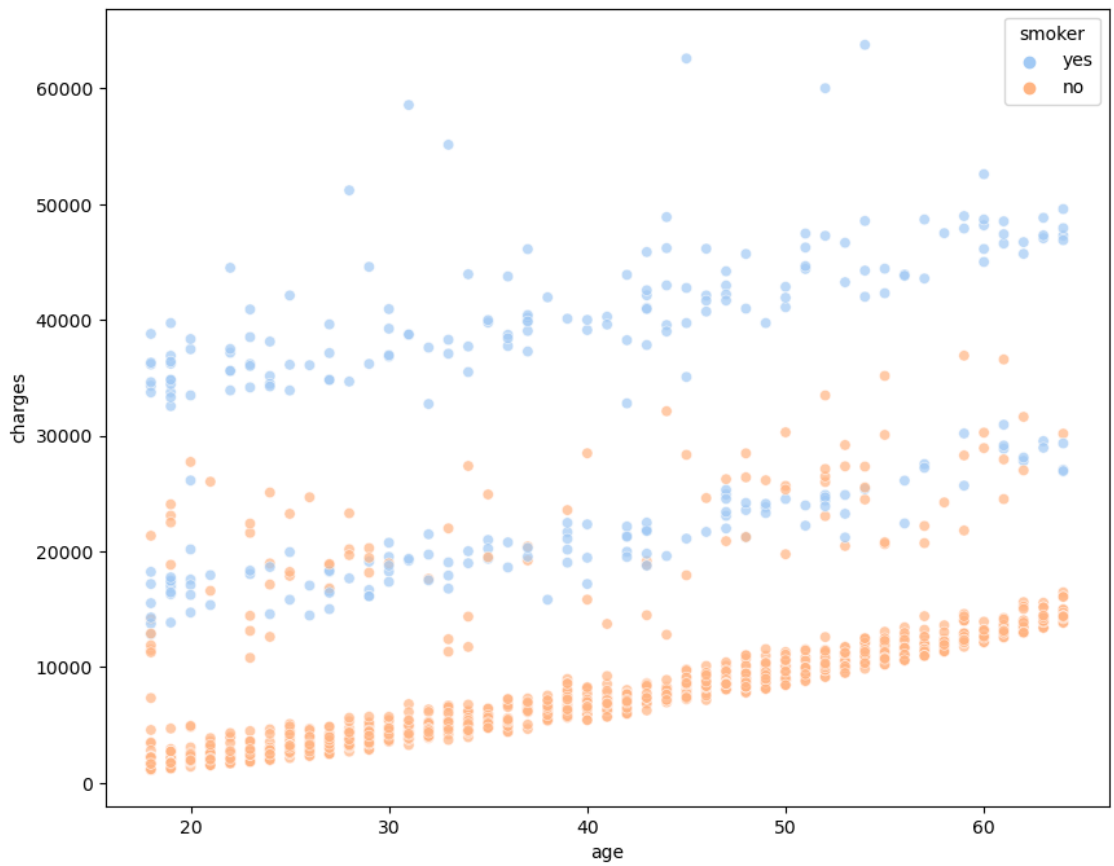


- 6.Check if the number of premium charges for smokers or non-smokers is increasing as they are aging

[49]:
```
#plotting the age vs charge graph
#hint draw a scatterplot age vs charges


plt.figure(figsize=(15,8))
sns.countplot(x='age',data=df)
plt.show()
```



[54]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,8))
sns.
↪scatterplot(data=df,x='age',y='charges',hue='smoker',palette='pastel',alpha=0.
↪7)
plt.xlabel('age')
plt.ylabel('charges')
plt.legend(title='smoker')
plt.show()
```


smoker
yes
no
60000
50000
40000
30000
20000
10000
0
charges
20
30
40
50
60
age

```
[ ]:
```